BEFORE THE NEW YORK STATE

PUBLIC SERVICE COMMISSION

Case 23-G-0225: Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

Case 23-G-0226: Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service.

DIRECT TESTIMONY OF SANE ENERGY PROJECT

September 1, 2023

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I. INTRODUCTIONS AND QUALIFICATIONS

Q. Please state your name and business address.

A: Kim Fraczek; 442 Lorimer St., Ste D - PMB 5064, Brooklyn, NY 11206

Q. What is your occupation?

A: Director, Sane Energy Project

Q. Have you previously testified before the Commission?

A: Yes.

- 19-G-0066 & 19-E-0065 Active Party in Con Edison's 2019 Rate Case
- 19-G-0309 & 19-G0310 Active Party National Grid's KEDNY KEDLI 2019 rate case.
- 19-G-0678 Active Party in the Public Service Commission ("PSC") Investigation of National Grid Denials of Service
- 20-G-0131 PSC's Motion for Gas Planning Procedures.
- 20-M-0266 PSC's Motion for the Effects of COVID-19 on Utility Service.
- 22-M-0429 PSC's Motion to Implement the Requirements of the Utility Thermal Energy Network and Jobs Act. ("UTENJA")
- 22-M-0149 PSC's Motion for Implementation of and Compliance with the Requirements and Targets of the Climate Leadership and Community Protection Act ("CLCPA").
- In addition to providing expert testimony and participating in settlement negotiations, Sane Energy Project ("Sane") has helped thousands of New Yorkers, including impacted

frontline community members, understand and engage in the rate case process and other regulatory proceedings before the PSC. Our information sessions and outreach events led to hundreds of people testifying at public hearings and submission of tens of thousands of written comments in numerous cases before the PSC on energy issues.

Q. What is Sane Energy Project?

A. Established in January of 2011, Sane Energy Project is a non-profit organization with a broad constituency of over 16,000 New Yorkers, notably inclusive of 12,000 individuals who are customers of National Grid Downstate within the KEDNY/KEDLI Territory.

Our core mission revolves around the replacement of fracked gas infrastructure with an unequivocally democratic and sustainable energy framework throughout New York State. Central to our approach is the principled integration of racial, social, and ecological justice into all our endeavors.

We actively engage in collaborative efforts with scientists, health professionals, industry experts, and communities profoundly impacted by energy infrastructure developments across the state. Our overarching objective is to establish the foundations for a renewable energy system guided by democratic principles, wherein communities possess the agency to shape decisions that directly influence their daily lives.

Our initiatives are underpinned by a dual focus: raising awareness regarding the consequential health and climate implications associated with fracked gas infrastructure and actively collaborating with communities to identify and dismantle obstacles obstructing the realization of community-owned renewable energy projects.

A testament to our commitment, we have extensively mobilized and empowered thousands of New Yorkers to effectively prepare and submit public commentary. This participation has been instrumental in influencing the permitting and regulatory processes pertaining to more than a dozen fracked gas infrastructure projects situated throughout the state of New York.

Q. Please summarize your expertise and background.

A: I have been an integral member of the Sane Energy Project team since its inception in 2011, serving in various capacities that underscore my commitment to the organization's mission, first taking on the role of volunteer coordinator for Sane Energy Project from 2011-2013. During this initial voluntary engagement, my focus was to address and rectify the concerning water and air quality issues precipitated by fracking infrastructure within my hometown in Pennsylvania. Furthermore, I was deeply committed to championing the cause of numerous communities across our region, nation, and indeed the world, who find themselves disproportionately affected and marginalized when confronted with the profit-driven machinations of the fossil fuel industry.

Prior to my formal employment with Sane Energy, my extensive professional background spanning over 12 years in corporate creative direction and digital rapid prototyping, I have overseen and managed multifaceted teams of designers and manufacturers located in diverse global hubs such as New York, Tokyo, Hong Kong, and Visakhapatnam, India.

My volunteer dedication along with my strong team skills was the impetus for my decisive career transition in 2013, when I left the corporate realm to establish Sane Energy Project as a cornerstone in New York State renewable energy policy advocacy.

The primary goal behind this transition was to foster the growth of Sane Energy Project into an institution capable of sharing essential educational and professional resources with frontline communities. These communities have consistently shouldered the burdens imposed upon them by corporate entities wielding disproportionate wealth and power.

In the pursuit of our mission, I have undertaken extensive travel throughout Pennsylvania and New York, working closely with communities, collaborating with esteemed scientists and medical professionals, and engaging in constructive dialogues with elected officials.

Our collective aim has been to effect changes in legislative, regulatory, and permitting frameworks that have, to date, exhibited leniency toward the fossil fuel industry, which has systematically compromised our fragile ecosystem, public health, and democratic principles.

Q: What is your purpose in submitting this testimony?

A: The objective of this testimony is to elucidate National Grid's shortcomings in formulating a proposal that duly considers the distinctive climate and health ramifications associated with fracked gas. Furthermore, it is imperative to highlight the mounting societal call for a transition away from fossil fuels. National Grid's proposal, which allocates substantial ratepayer funds to Storm Hardening, concurrently entails investments exceeding hundreds of millions of dollars in fossil fuel infrastructure. Regrettably, such an approach not only perpetuates climate change but also exacerbates the intensity of future storms.

II. GAS ISSUES

Introduction And Background:

In light of the pressing challenges posed by climate change and the imperative of adhering to the Climate Leadership and Community Protection Act (CLCPA), as well as evaluating proposals related to fracked gas infrastructure, it is essential to scrutinize National Grid's request for $3.8 billion in billing to New Yorkers over the next four years. This proposal, intended to facilitate the expansion and extension of the gas network, raises serious concerns about its environmental and ethical responsibility.

We urge the New York State Department of Public Service (NYSDPS or “Staff”), The City of New York (“NYC”), and The Public Service Commissioners (“PSC”) to uphold both the letter and spirit of our state's laws. In the mission of the PSC, we find a commitment to "stimulate effective competitive markets for clean, renewable, and distributed energy resources that benefit New York consumers, as well as product and service innovations."

National Grid's plan for gas network expansion and extension is at odds with the CLCPA, failing to align with the mandated goals of transitioning away from fossil fuels by 2050, while directing 40% of state climate and energy funding toward historically disadvantaged communities.

The proposal's core flaw lies in its attempt to expand and prolong aging and leak-prone infrastructure over the next four years, falsely portraying this action as emissions reduction, only to witness these same structures continue to deteriorate and emit past 2050. Such an approach is incompatible with the legally binding emissions reduction standards and the overarching objective of eliminating New York's dependence on fossil fuels by mid-century.

Moreover, National Grid's endorsement of fossil fuel expansion demonstrates a stark disregard for

the profound implications of climate change and the injustices experienced by New Yorkers in recent times. In the year 2023, we confronted a summer marked by unprecedented temperatures, officially recorded as the hottest year on a global scale, as corroborated by data from authoritative sources such as NOAA and NASA[1]. During this period, the infiltration of Canadian wildfire smoke, driven by climatic shifts, enveloped National Grid's service territory, resulting in the closure of educational institutions and a surge in emergency room visits, particularly among hundreds of individuals afflicted by asthma.

Notably, a significant portion of National Grid's service territory constitutes an Urban Heat Island (UHI), typified by a densely paved urban environment that retains heat. Among 44 U.S. cities assessed, New York ranks the highest in the UHI Index[2]. This phenomenon disproportionately affects low-income communities of color, whose residents face elevated risks of mortality during and immediately following extreme heat events. An empirical analysis conducted by NYC employing the Heat Vulnerability Index (HVI) reveals that certain neighborhoods are particularly vulnerable to heat-related mortality, and it is crucial to underscore that these disparities are rooted in historical and contemporary racism.

Given that one of National Grid's most substantial investment requests pertains to an area proximate to 2,000 predominantly Black, low-income residents, we assert the compelling case for the retirement of the 287 Maspeth facility, or the “Greenpoint Energy Center”.

[1] https://www.nasa.gov/press-release/nasa-clocks-july-2023-as-hottest-month-on-record-ever-since-1880
[2] https://a816-dohbesp.nyc.gov/IndicatorPublic/beta/key-topics/climatehealth/hvi/#disparities

Climate Change and overall compliance with the Climate Leadership and Community Protection Act ("CLCPA")

Q: What kinds of gas capital expenses is National Grid proposing?

A: National Grid is proposing a range of capital expenditures aimed at replacing and expanding Leak Prone Pipelines (LPP) and other fracked gas infrastructure. These initiatives fall under the umbrella of the Proactive Main Replacement Program, Gas System Reinforcement Program, and Customer Connections.

Q: Does National Grid's project proposals to limit methane address the issue?

A: No, they do not. While the identification and rectification of existing leaks are certainly vital steps, the technologies proposed for leak detection do not adequately address the broader concern of upstream emissions, particularly when viewed in conjunction with pipeline and infrastructure enhancements that are poised to increase overall gas consumption.

Q: What actions should National Grid undertake instead of investing in gas capital projects for pipeline replacement and customer conversion?

A: Rather than allocating new ratepayer funds toward the expansion or extension of gas infrastructure, National Grid should pivot towards more sustainable alternatives. We strongly advocate for an aggressive transition of the gas system to Thermal Energy Networks. Please see a

more comprehensive overview in the next section. Additionally, a concurrent intensification of efforts in gas demand reduction through comprehensive Energy Efficiency programs. These programs should not only target energy conservation but also address ancillary issues such as mold remediation and lead paint removal.

Q: Should National Grid announce they no longer need the expansion of the Iroquois pipeline in its Expansion by Compression project (EXC) in order to comply with the CLCPA?

A: Yes, National Grid should unequivocally cease pursuing support for the expansion project in Greene County (Athens Compressor Station, 915 Schoharie Turnpike, Athens, NY 12015) and Dutchess County in New York (Dover Compressor Station, 186 Dover Furnace Road, Dover Plains, NY 12522), as well as in neighboring Brookfield and Milford, Connecticut.

It is essential to recognize that emissions do not adhere to artificial borders or boundaries, and such expansion projects should be reevaluated in light of the Climate Leadership and Community Protection Act's (CLCPA) mandate to account for both upstream and downstream emissions.

Q: How have local residents in rural New York appealed to New York State to reject the Iroquois EXC Project?

A: Residents residing in the rural Harlem and Hudson Valleys, who stand to be directly impacted by the proposed expansion, have taken action by submitting a total of 2,388[3] comments to the New York Department of Environmental Conservation regarding permit applications 3-1326-

[3] https://actionnetwork.org/letters/tell-hochul-dec-deny-the-air-permits-for-iroquois-pipeline-compressors?clear_id=true

00211/00001 (Dover Compressor Station) and 4-1922-00049/00004 (Athens Compressor Station). These comments collectively advocate for the denial of the Article 19, Air State Facility permit requests associated with these projects.

Q: What is the Iroquois Pipeline?

A: The Iroquois Pipeline is an aging, 414-mile-long pipeline that has been in operation for 37 years. It traverses from Canada through the state of New York, passes through Connecticut, and re-enters New York under the Long Island Sound before reaching the National Grid KEDLI territory.

Q. What is the scandal surrounding National Grid and the Iroquois EXC project pending before the NYS Department of Environmental Conservation?

A: The controversy surrounding National Grid and the Iroquois EXC project centers on the circumstances that led to its proposal. In 2019, then-Governor Andrew Cuomo mandated National Grid to explore alternatives to the highly contested[4] and ultimately rejected Williams NESE pipeline project. National Grid's actions during this period raised significant concerns and led to allegations of scandal. Notably, National Grid's decision to reject gas hookups just prior to the winter of 2019-2020, citing a lack of gas supply, was viewed by many as a strategic move to manipulate the New York government into approving an unnecessary pipeline. Such an approval would have been financially advantageous for National Grid. These actions were met with

[4] https://350.org/wp-content/uploads/2019/03/Stop_Williams_False_Demand.pdf

widespread criticism and calls for greater transparency and accountability in energy infrastructure decisions.

GAS INFRASTRUCTURE OPERATIONS AND SUPPLY PANEL

Leak Prone Pipe

Q: What does Sane Energy Project propose should happen with the Leak Prone Pipeline replacement program?

A. Our proposal strongly advocates for the retirement of Leak Prone Pipelines (LPP) instead of their replacement, and it emphasizes the importance of transitioning customers to non-gas alternatives as part of a progressive and forward-thinking strategy. In National Grid's SANE-6[5] response, these provided maps and plans presented, within the KEDNY and KEDLI territories, leave little room for innovation. It appears as though the entire service territory, especially in Brooklyn, is marked for the replacement of gas pipelines. This approach is outdated and represents an imprudent allocation of the public's financial resources.

Furthermore, I would like to draw attention to our recommendations outlined below, particularly concerning the implementation of Thermal Energy Networks, which offer a more sustainable and forward-looking solution.

[5] SANE-EXHIBIT-1 SANE-06 Response.pdf

Greenpoint Energy Center

Q: What does Sane Energy Project propose should happen to the Greenpoint Energy Center?

A: Considering that the most substantial allocation of funds, totaling an impressive $448 million, pertains to the Greenpoint Energy Center located at 287 Maspeth Ave, and recognizing that National Grid commands the largest parcel of land in North Brooklyn—an area in dire need of green space to combat the urban heat island effect, as previously discussed—we strongly advocate for the establishment of a well-defined and measurable plan. This plan should set a clear milestone for the complete retirement of all gas infrastructure at the Greenpoint Energy Center by 2030.

It is pertinent to note that community members within our jurisdiction have invested significant volunteer efforts to transform previously polluting industrial spaces into remediation centers and accessible public areas. Notably, the baseball fields at 287 Maspeth have regrettably fallen into disuse and disrepair due to elevated toxicity levels, further exacerbated by the recent discovery of the Meeker Avenue plumes. In light of these circumstances, it becomes paramount for National Grid to adopt a stewardship role in relation to a facility occupying a landmass equivalent to the entirety of the available green space north of Flushing Avenue in Brooklyn. To put this into perspective, National Grid's 287 Maspeth facility spans approximately 120 acres, while the cumulative total of all green spaces in North Brooklyn amounts to approximately 118 acres. This incongruity underscores the pressing need to recalibrate priorities toward fostering a healthier and more sustainable renewable future for the community.

Indeed, there is no more compelling means for National Grid to demonstrate its sincere commitment to the well-being and safety of its customers than by redirecting its investments towards sustainable and renewable energy sources. Such a shift not only underscores the company's recognition of the limitations and risks associated with fossil fuel investments but also aligns with the imperative of safeguarding the long-term health and vitality of the communities it serves. By embracing cleaner and more sustainable energy solutions, National Grid can exemplify its dedication to both its customers and the broader cause of environmental responsibility. By expanding green space needed in the service territory to address the Urban Heat Index, members of the community living near the facility share concerns, and have recommendations for National Grid.[6]

Q: Why is an investment of just under $488 million considered an abominable expenditure for a facility that sees limited use?

National Grid states in SANE-02 Response[7]

> "Without the Greenpoint LNG plant, several significantly more expensive pipeline expansions would be needed to meet the current demand on the system under not only existing peak day and peak hour conditions, but starting when average temperatures fall below 15°F, which typically occurs every winter. The Greenpoint LNG plant is also one of the Company's most important assets to ensure reliable service for gas customers, as it serves as a backup should upstream pipeline supplies be interrupted during the winter heating season."

[6] SANE-EXHIBIT-2-Kevin LaCherra Interview About Greenpoint Energy Center (1).pdf
[7] SANE-EXHIBIT-3-SANE-02 Response.pdf

The justification for deeming an investment of nearly $488 million as highly objectionable for a facility with minimal utilization stems from a critical examination of the circumstances. Sane Energy Project's past advocacy led to the commissioning of an independent review regarding the necessity of Vaporizers 13 and 14 at the Greenpoint Energy Center during National Grid's 2019 Rate Case. This impartial assessment unequivocally concluded that such costly infrastructure was, in fact, superfluous.

In light of these findings, we strongly recommend that both the Public Service Commission (PSC) and its staff initiate a comprehensive evaluation of the entirety of the Greenpoint Energy Center. If this facility, sprawling across 120 acres, is proclaimed to be "one of the Company's most important assets" but serves a functional purpose only during a few cold days each year, it raises concerns about the responsible allocation of New Yorkers' funds within the KEDNY service territory.

It is fundamentally untenable to justify the expenditure of nearly half a billion dollars to extend the lifespan of this facility, particularly in the context of our warming climate, significantly influenced by such fossil fuel investments.

Furthermore, the United States Army Corps of Engineers has identified[8] the specific area of Newtown Creek as highly susceptible to the adverse effects of anticipated rising waters in this floodplain. They have proposed the construction of an 18,000-foot barrier to manage surge waters,

[8] https://www.nan.usace.army.mil/Portals/37/docs/civilworks/projects/ny/coast/NYNJHAT/NYNJHAT%20Interim%20Report%20-%20Main%20Report%20Feb%202019.pdf?ver=2019-02-19-165223-023

although the appropriateness of this measure remains a subject of ongoing debate regarding sea-level rise.

Nonetheless, it is undeniably evident that sea-level rise is a looming threat, and the location occupied by National Grid is in close proximity to this vulnerable area. In light of these circumstances, it is deeply troubling that National Grid is requesting nearly half a billion dollars of public funds to fortify precisely the kind of infrastructure that contributes to the exacerbation of sea-level rise.

Furthermore, while National Grid contends in response to SANE-3[9] that continued investment in the Salt Water Pump House is essential for critical fire protection equipment, we wish to emphasize our objection to the ongoing operation of the LNG (liquefied natural gas) facility. While we fully endorse adherence to FDNY (New York City Fire Department) code requirements, we firmly assert that there is no valid justification in the 21st century to persist in the utilization of such hazardous and flammable materials. The financial burden imposed by the maintenance of this facility, which poses both environmental and safety risks, is exorbitant and unjustifiable.

This underscores the urgency of proactively retiring fossil fuels on this property in a swift, orderly and safe way.

[9] SANE-EXHIBIT-4-SANE-03 Response.pdf

Q: How does Sane Energy Project envision a more rational alternative to constructing new LNG infrastructure, such as a Vapor Suppression System at the Greenpoint Energy Center?

A: In Sane-7 response[10], National Grid's proposition of the Vapor Suppression System project is that it is designed to enhance safety by installing a passive system that minimizes the amount of vapor released in the event of an LNG spill. But, it is crucial to consider the broader context.

We wish to draw National Grid's attention to LNG safety data provided by your colleagues at Philadelphia Gas Works[11], which underscores the hazardous nature of LNG vapors. These vapors have been shown to have detrimental effects on the central nervous system and can act as asphyxiants, potentially leading to loss of consciousness, severe injuries, or even fatalities by displacing air, resulting in insufficient oxygen to support life.

It is perplexing that National Grid continues to perceive LNG as a rational choice in the 21st century, particularly considering its intention to store substantial quantities of LNG in close proximity to a residential neighborhood and beneath one of the busiest arteries in New York City. Our perspective is that by agreeing to a retirement plan for this facility and conceiving a comprehensive strategy for the safe extraction of flammable fuels for energy production, we could establish a facility that fulfills energy requirements, ensures public safety, and mitigates climate impacts. Such a facility could serve as a model for other communities facing similar challenges but lacking the progressive and engaged governance found in New York.

We are presented with a remarkable opportunity in New York to set an example for others to follow.

[10] SANE-EXHIBIT-5-SANE-07 Response.pdf
[11] https://www.pgworks.com/uploads/pdfs/LNGSafetyData.pdf

Q: Should National Grid retire Vaporizers 7 and 8 at the Greenpoint Energy Center?

A: Yes, National Grid should indeed retire Vaporizers 7 and 8 at the Greenpoint Energy Center. Instead of channeling additional funds into aging 50-year-old infrastructure, which seems to be veiled under the guise of capital improvements and the use of safety and reliability as rhetorical devices to sway the Public Service Commission (PSC), it would be more prudent to retire these units. The considerable financial outlay requested for this project should be thoroughly scrutinized in light of the broader context and the potential for more sustainable and responsible alternatives.

Q: Should any pipelines in Eastern Long Island, such as the KEDLI Southampton-Transmission-Main, be retired in favor of Thermal Energy Networks?

A: Yes, it is prudent to consider retiring pipelines in Eastern Long Island, including the KEDLI Southampton-Transmission-Main. Many residents in this area still rely on oil or propane tanks for their energy needs. Therefore, it would be a responsible course of action to transition these residents directly to renewable energy sources rather than expanding the gas pipeline system. It is worth noting that there was significant opposition to the South Suffolk pipeline during the 2019 National Grid rate case. However, it appears that National Grid has continued to proceed with its plans despite objections, maintaining the status quo instead of actively seeking more sustainable alternatives.

V. THERMAL ENERGY NETWORKS

Thermal Energy Networks for alternative energy options to customers seeking new connections

Q. What is the relevance of National Grid Utility Thermal Energy Network Pilot Proposal, Case 22-M-0429 to this rate case, Case 23-00902/23-G-0225?

A. National Grid/KEDNY's Utility Thermal Energy Network Pilot Proposal in Case 22-M-0249 ("UTEN Pilot") is relevant to this case, Case 23-00902/23-G-0225, given the request in the latter for recovery of costs of the FTEs associated with the proposed Vandalia Avenue Thermal Energy Network (TEN) pilot project. More broadly, thermal energy networks are an essential, carbon-free, low-cost, non-pipe alternative to the gas infrastructure. In its filings, the Company largely ignores the potential of TENs to reduce demand for gas for heating and hot water, as well as electricity for cooling, preferring instead to seek a massive funding increase to extend the lifetime and reach of its fracked gas infrastructure.

Q. What costs does the Company seek to recover for Case 22-M-0249 ("UTEN Pilot") in this case, Case 23-00902/23-G-0225?

A. In the Gas Infrastructure and Operations Panel (GIOP), the Company states that: On January 9, 2023, the Company submitted a thermal energy network pilot proposal in Case 22-M-0249 ("UTEN Pilot"). The Company is seeking approval for cost recovery of the UTEN Pilot in Case 22-M-0429 and not in this proceeding. In this rate case the Company is seeking five FTE positions

to support the successful implementation of the UTEN Pilot. Of the five FTEs requested, the Company needs: (1) three FTEs in gas asset engineering to design and develop networked systems, work closely with contractors, and conduct outreach to partners and customers; (2) one FTE for a maintenance mechanic responsible for installation, inspection, and repair of the installed networked system(s); (3) 0.5 FTE for a lead program manager to deliver the program and prepare reports and analytics; and (4) 0.5 FTE for a lead project manager who will have responsibility for the networked thermal energy project from project approval, implementation, development, execution, and closure.[12]

Q. Is the request for UTEN FTEs for Case 22-M-0249 ("UTEN Pilot") appropriate?

A. No. While we strongly support in principle the design and construction of UTENs by the Company, and specifically endorse the implementation of well-designed UTENs in Disadvantaged Communities (DACs) and NYCHA facilities, we question the specification of three FTE "gas asset engineers" for KEDNY's proposed UTEN project. UTEN projects need engineers with expertise in thermal and hydronic systems and multifamily and commercial building HVAC. In job postings for gas asset engineers, the Company clearly states that these positions are for specialists in gas distribution systems, gas operations, and gas facilities construction,[13] expertise which is not directly relevant to the UTEN pilot but is instead relevant to the leak-prone gas line replacement and expansion effort that is the primary focus of this rate case. Moreover, assigning these gas asset engineers the responsibility for conducting "outreach to partners and customers"[14] is neither an

[12] **SANE-EXHIBIT-6-Gas Infrastructure and Operations Panel.pdf Page 87 of the GIOP Panel**
[13] SANE-EXHIBIT-13-National Grid Gas Asset Engineer Job Description from GEEBO August 2023.pdf
[14] **SANE-EXHIBIT-6-Gas Infrastructure and Operations Panel.pdfPage 87 of the GIOP Panel, line 7**

appropriate use of their skills nor an effective strategy for reaching out to and informing stakeholders about UTENs. Seeking cost recovery for gas asset engineers under cover of the proposed KEDNY UTEN is inappropriate and could suggest that the Company is misrepresenting how it would use personnel ostensibly hired for the UTEN pilot.

Q. What is KEDNY's Utility Thermal Energy Network Pilot Proposal?

A. In its January 9, 2023 submission and in its May 31, 2023 response to comments from the Department of Public Service, KEDNY proposed what can best be described as heat-only, borefield-based district geothermal system for the New York City Housing Authority (NYCHA) buildings buildings located at 17, 47, and 77 Vandalia Avenue in Brooklyn. Sane Energy Project found the design to be insufficient and incomplete.[15]

Q. In what ways was National Grid's Utility Thermal Energy Network Pilot Proposal insufficient and incomplete?

A: Sane Energy Project outlined some of the issues in comments submitted in response to the Company's proposal, noting that:

> KEDNY has not proposed a real TEN. Rather, it has simply described a large geothermal system which would capture thermal energy from boreholes. There is no suggestion of thermal energy capture and recirculation in the superficial schematic the company

[15] SANE-EXHIBIT07-2023 01 09 KEDNY UTEN proposal.pdf + SANE-EXHIBIT-16-2023 05 31 Nat Grid response to commennts.pdf

provided, and the text mentions only two stores in nearby strip malls as possible sources of heat capture. The fundamental balancing principle of thermal energy networks that involve diverse buildings and needs and a variety of heat sources and sinks are completely absent from the design. The proposal acknowledges the great value of TENs: "the networked system enables thermal energy sharing among buildings by leveraging the diversity of concurrent heating and cooling dominated loads" (11), but it does not incorporate that core element into the Brooklyn plan.

Equally egregious is the plan to provide only heating, not cooling. This would leave at least half of the potential of the heat pumps and the pipelines circulating fluid unused, a tremendous waste of the financial investment and the technology. Residents would have to rely on window air conditioning units to make their homes tolerable in our longer, hotter summers. External equipment like window ACs deteriorate rapidly from exposure compared to underground geothermal pipes and indoor heat pumps, adding maintenance and replacement costs. And the cost of electricity for running those ACs, whether borne by the tenants or by NYCHA and the taxpayers, would be a significant added expense that would also decrease the thermal and economic value of the System.[16]

The heat-only provision also violates a core principle guiding the PSC: resilience. By not using efficient thermal energy networks for cooling, the buildings' tenants will, as noted above, rely on electric air conditioners during the summer. This will drive up electricity demand and increase the likelihood of blackouts. Designing and building a

[16] SANE-EXHIBIT-8-2023 04 05 Sane Energy Comments on KEDNY proposal.pdf pp 2-3

highly efficient system that provides cooling would sharply cut electricity use, increase reliability, and also store underground heat for the next winter. The proposal notes that using electricity for cooling also results in GHG emissions from electricity generation; that would be eliminated if a proper TEN were designed and built.[17]

Q. Is the Vandalia Avenue NYCHA complex a suitable site for a UTEN?

A. Yes, but only if the thermal energy network is properly designed, which the KEDNY proposal as written and later revised is not. In April, when we wrote our initial comment, we thought the Vandalia Avenue NYCHA facility proposal, with its reliance on a very expensive field of 499 geothermal boreholes and the barriers to including cooling, was simply a poor choice for a UTEN site. Sane's comment therefore urged "the PSC to reject the KEDNY's proposal as written, and demand that it be modified for another location such that the pilot would incorporate all of the benefits of thermal energy networks."[18]

However, because we strongly support the goal of providing efficient, healthy, carbon-free, renewable, low-cost heating, cooling, and hot water to NYCHA residents through thermal energy networks, and because the pilot projects mandated by UTENJA should test and ideally demonstrate that TENs can be designed and constructed for a wide range of building types and site conditions, Sane Energy Project contracted at our cost with Egg Geo, a highly qualified, NYSERDA-registered external peer reviewer specializing in TENs, to assess the feasibility of the pilot as presented and to propose possible improvements. The initial report is entitled "Scoping Study/Peer Review is of proposed U-TEN for Brooklyn NYCHA." That study found serious, probably disqualifying, inadequacies and unnecessarily high costs in the KEDNY UTEN proposal. It also

[17] SANE-EXHIBIT-8-2023 04 05 Sane Energy Comments on KEDNY proposal.pdf p. 3
[18] SANE-EXHIBIT-8-2023 04 05 Sane Energy Comments on KEDNY proposal.pdf p.4

identified new design elements that would enable construction of an exemplary thermal energy network at the Vandalia Avenue site using adjacent thermal resources rather than drilling 499 boreholes, saving millions of dollars. Those resources include a New York City Transit Authority (NYCTA) subway dewatering facility and Wastewater Energy Transfer (WET) from an NYC Department of Environmental Protection wastewater treatment plant. The Egg Geo study posits that cooling and water heating can be included at less that $40 million–roughly the same cost as the heating-only system proposed by the Company.[19]

The Egg Geo Survey also notes that, while a thermal energy network could reduce summer electricity spikes driven by conventional air conditioning, implementing TEN pilots is also a chance to do necessary, comprehensive weatherization and electrification upgrades. As the Egg Geo study states:

> Finally, the opportunity to properly upgrade the electrical system in these buildings has come. If it is determined that the building needs 220-volt service in each apartment, it should be done in a pilot project such as this. *This is not the time to go with an inferior solution.* As the mandate for building electrification and decarbonization is here to stay, it would be better to realize the best upgrade solutions[20]

Q. Given the weaknesses of KEDNY's proposed Vandalia Avenue TEN, should the utility propose additional thermal energy network pilots?

A. Yes. It is regrettable that the Company has presented only one proposed UTEN pilot project, while seeking cost recovery and extreme rate hikes to replace and extend the life of its methane-

[19] SANE-EXHIBIT-9-Egg Geo Scoping Study UTEN 22-M-0429 F.pdf p.14
[20] SANE-EXHIBIT-9-Egg Geo Scoping Study UTEN 22-M-0429 F.pdf p. 18

leaking pipes in order to ensure reliable gas service for heating and other needs in the face of increased demand. Designing, deploying, and scaling up thermal energy networks could significantly cut demand for both gas and electricity within three years, if the utilities and other potential developers take action. The UTENJA requires each of the seven largest utilities to develop between one and five TEN pilots.[21] We believe that the Company should seize the opportunity to properly design and build more TENs. It is vital for the people of New York, the global climate, and, given the mandates of the CLCPA and the momentum away from gas in the building sector, KEDNY itself. The Company has an extraordinary opportunity to change its business model from obsolete reliance on fossil fuels to renewable, affordable thermal energy networks. That shift would allow the Company to utilize many of its existing assets and position itself for long-term success.

Some of KEDNY's utility peers recognize the opportunity that UTENJA presents. Con Edison in particular is notable for proposing three TEN pilots and two feasibility studies.[22] KEDNY should identify four more possible sites as allowed and encouraged by the Utility Thermal Energy and Job Act (UTENJA), and beyond that develop feasibility studies for a pipeline of additional UTENs. The value of pilot projects lies in designing and testing varied applications of the technology in diverse settings, with different commercial, residential, and government buildings and facilities, diverse sources and sinks including wastewater and flood management, subways, data centers, etc. We urge National Grid to cast off its antiquated commitment to climate-killing fracked gas systems and step into the twenty-first century. Developing and implementing first-rate thermal energy

[21] SANE-EXHIBIT-10-UTENJA Bill 2022.pdf
[22] SANE-EXHIBIT-11-2023 05 19 ConEd Pilot Proposal Supplement.pdf

networks in densely populated Brooklyn, with its varied building stock and proximity to waterways and other infrastructure, is the gateway to the Company's future success.

VI. PROCEDURAL RECOMMENDATIONS FOR RATE CASE PROCESS

A. Public Hearing Advertising and Company Advertising and Community Board Outreach 2019 and 2023

Q: Is there a discrepancy between how National Grid advertises for programs and announcements they want to promote and how National Grid advertises for public hearings and Community input?

A: Yes, there is a discernible incongruity in how National Grid handles advertising for its programs and initiatives compared to its approach to public hearings and community input. Despite being a corporation that boasts billions of dollars in profits[23], it appears that National Grid has the capability to allocate significant resources to promoting public participating processes with professional graphic design and public relations teams in order to effectively engage customers in various initiatives.

When it comes to public hearings and soliciting community input on projects that directly affect people's lives and the energy system, a professional and attractive level of engagement and visual appeal is notably absent. While National Grid may claim to have advertised for public hearings, the advertisements lack the visual and informational quality that their programs demonstrate they are capable of.

[23] National Grid 2022/23 Full Year Results Statement

It is worth highlighting that in the 2019 National Grid rate case, the company asserted that it had conducted outreach to Community Board 1. However, Community Board 1 responded that they had not been contacted. In response to this discrepancy, community organizations and elected officials had to take the initiative to organize a press conference, urging National Grid to address concerns related to the Metropolitan Reliability Infrastructure Project, also known as the North Brooklyn pipeline at the Community Board 1 Brooklyn meeting on January 14, 2020.

This pattern of minimal effort in notifying the public about hearings and lacking meaningful engagement in the public process is evident and raises concerns about the transparency and inclusivity of National Grid's practices.

Additionally, it is essential to highlight the impracticality and inaccessibility of relying on staff-published schedules within the preceding docket as a means of notifying the public. Navigating the New York State Department of Public Service (DPS) website is a daunting task for the average person, and finding information about public hearings buried within a PDF document is far from acceptable. It is inconceivable that the average working person, who dutifully pays their National Grid bill, would have the means or the know-how to discover and participate in a public hearing merely by sifting through documents in a docket.

In conclusion, it is abundantly clear that state agencies, the company, and other stakeholders are disproportionately reliant on organizations like Sane Energy Project to expend their limited resources in genuine efforts to engage the public. These organizations are tasked with preparing and educating the public about the intricacies of proposed projects and how to participate effectively in public hearings. It is disingenuous to assert that an honest and adequate effort has been made to involve the public in the process of providing testimony and shaping public services

that profoundly impact their everyday lives. National Grid's claim of putting forth an honest effort is deeply troubling and appears to lack authenticity.

Furthermore, it is worth noting that state agencies, city agencies, and corporations involved in rate cases have significant resources derived from tax revenue and ratepayer funds. Despite these resources, they still rely heavily on grassroots and community groups to conduct actual outreach and engage the public. By any reasonable standard, this practice represents a profound injustice and a flagrant display of dishonesty.

Q: Are Interrogatory Discovery Questions answered with good-faith attempts to hear the complaints of community groups?

A: No, unfortunately, it appears that Interrogatory Discovery Questions, especially in the case of SANE-5[24], are not answered with genuine good-faith efforts to address the concerns of community groups. The responses often come across as superficial and lacking in substantive engagement. They resemble a student's attempt to submit a paper filled with fluff in an effort to create the illusion of research or understanding. In particular, the response to SANE-5 appears to resemble a text-banking script rather than a sincere effort to engage with the issues raised by the community group. Additionally, the claim that everyone has equal access to information about alternatives to gas is not entirely accurate and may be viewed as an attempt to fulfill the minimum requirements rather than genuinely addressing the concerns of the community. It is imperative that responses to Interrogatory Discovery Questions demonstrate a sincere effort to listen to and understand the complaints and perspectives of community groups.

[24] SANE-EXHIBIT-12-SANE-05 Response.pdf

Q: Does Chair Rory Christian's assessment of empowering the public translate to the Department of Public Service Process of engagement in proceedings?

A: Unfortunately, it appears that Chair Rory Christian's inspiring words[25], as mentioned in the Fortnightly article, do not translate into tangible actions within the Department of Public Service's engagement processes. While Chair Christian's words may inspire hope and convey a commitment to empowering the public, the reality on the ground suggests otherwise.

In practice, it often falls upon the public and small non-profit organizations to tirelessly invest significant time and effort, often working late into the night, seven days a week, to participate and shape a better world. This commitment stands in stark contrast to the well-resourced parties involved in proceedings, who do not seem to align with Chair Christian's stated aspirations of empowering the public. The public engagement process should be a collaborative and inclusive effort, but the current reality does not always reflect these principles:

> The swell of public interest in energy and environmental matters has surged into the forefront of our nation's consciousness. Record-breaking weather, coupled with increasingly frequent extreme climate events, has ignited a desire within the public for a change from the status quo. Coupled with that desire for change, is a need to be heard and to shape the actions that follow.
>
> Public Service Commissions and Boards of Public Utilities, vested with the power to drive change, can embrace this burgeoning interest, and align actions and policies to expedite desired outcomes. We must explore innovative avenues that

[25] https://www.fortnightly.com/node/42786

facilitate engagement, embracing equity and inclusivity, as we educate, inform, and empower public engagement in our vital work.

The public, accompanied by a myriad of new stakeholders, are engaging public utility commissions and boards at unprecedented levels. The shift from in-person to remote meetings compelled by the constraints of the pandemic, has widened the spectrum of participation.

The evidence is overwhelming: What was once a mere trickle of public comments in significant rate cases has transformed into a torrent. A decade ago, a major rate case in New York garnered no more than six hundred public comments, whereas recent proceedings have received several thousand.

Rate cases, the lifeblood of Commissions, present a formidable challenge. While any party can participate, the path to meaningful engagement is fraught with obstacles that perplex the uninitiated and often test the patience of seasoned participants. Some maintain the view that newcomers hinder rather than enhance the overall discussion. That could not be further from the truth.